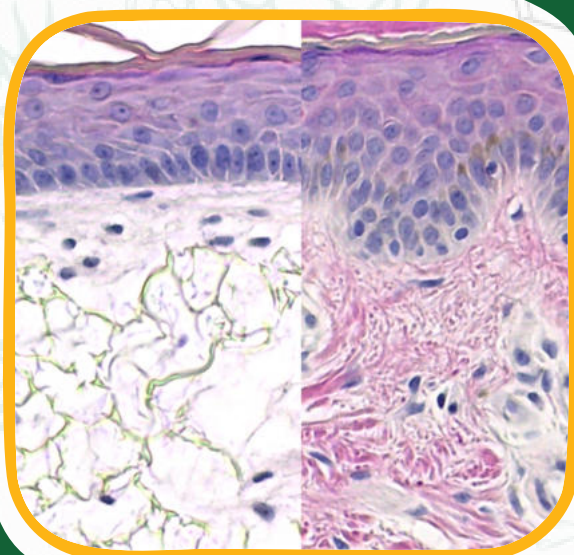




UNIT NO.4

CELL AND TISSUES



DEFINITIONS

CELL:

It is the fundamental unit of structural and functional in all living organism.

PLASMODESMATA:

The opening in the cell wall are plasmodesmata which contain strands of cytoplasm that connect adjacent cell.

PLASMOLYSIS:

It is the process in which cytoplasm of the cell shrinks due to loss of water when placed in a hypertonic condition.

NUCLEOPLASM:

The nucleus is filled a gel like granular fluid called nucleoplasm which has nucleoli and chromatin network.

DETOXIFICATION:

It is the physiological or medicinal removal of toxic substances from a living organism.

CARTILAGE:

It is a dense connective tissue that protects joints and bones.

CYCLOSIS:

The streaming movement of the cytoplasm is called cyclosis.

CISTERNAE:

Golgi body consist of a stack of flat membrane bound sacs called cisternae.

EXOCYTOSIS:

It is the process by which cell get rid of large molecules of waste products or toxin.

VESICLES:

It is small structure within a cell, consisting of liquid enclosed by a lipid layer.

SHORT/DETAILED ANSWER & QUESTIONS

Q. What is a microscope? Who discovered first microscope.

Ans. **MICROSCOPE:**

Microscope are instruments designed to produce magnified visual or photographic images of objects too small to be seen with the naked eye.

DISCOVERY OF MICROSCOPE:

The first compound microscope was developed by Zacharias Janssen in Holland 1590. He designed was based on simple hollow tube with lenses at each end. Its magnification ranged 3X to 9X. Later on, Robert Hook improved his version for compound microscope.

Q. What is microscopy? How many parameters are important for microscopy?

Ans. **MICROSCOPY:**

The use of microscope is called microscopy.

IMPORTANT PARAMETRS OF MICROSCOPY:

The two important parameters of microscopy are:

1. Magnification
2. Resolution

1. MAGNIFICATION:

The enlargement of an image is called magnification.

2. RESOLUTION:

It is defined as the smallest distance between two points on a specimen that can still be distinguished as two objects.

Q. How many types of microscopes. Explain the working principle of each of them.

Ans. **TYPES OF MICROSCOPES:**

There are two microscopes that are used in microscopy.

- Light microscope (LM)
- Electron microscope (EM)

LIGHT MICROSCOPE:

DEFINITION:

Microscope which uses visible light as a source of illumination and consists of lenses which magnify the light reflected from the object is called light microscope.

WORKING:

- A light microscope uses visible light as a source of illumination.
- A source of light such as mirror illuminates the object to be seen.
- The reflected light passes through a biconvex lens called the objective lens.
- The objective lens produces a magnified image of the object.
- This size of image is further enhanced by the lens located at the eyepiece of microscope.
- The magnifying power of light microscope is limited to 2000 times.
- Its resolution power has a very small value of 0.2 micro-meters.

ELECTRON MICROSCOPE:

Microscopes that produce an image of a specimen by using a beam of electrons is called electron microscope

WORKING:

- It uses a supply of electron beam as a source of illumination.
- It is equipped with electromagnetic lenses to focus the electronic beam on to the object.
- Photographic plate or screen is used to obtain the image.

RESOLUTION OF ELECTRON MICROSCOPE:

Electron microscope has a resolution as small as 0.2 nanometer (nm).

MAGNIFICATION OF ELECTRON MICROSCOPE:

Electron microscope has a magnification up to 250,000 times.

USES OF ELECTRON MICROSCOPE:

Electron microscopes can be used to examine not just whole cells, but also the sub-cellular structures and compartments within them.

LIMITATION OF ELECTRON MICROSCOPE:

A live cell cannot be imaged by electron microscope.

TYPES OF ELECTRON MICROSCOPE:

There are two major types of electron microscopes:

- Transmission electron microscope (TEM)
- Scanning electron microscope (SEM)

1. TRANSMISSION ELECTRON MICROSCOPE (TEM)

Transmission electron microscopy (TEM) is used to obtain detailed images of internal structure of cells. The specimen is cut into extremely thin slices before imaging, electron beam passes through the slice rather than skimming over its surface.

2. SCANNING ELECTRON MICROSCOPE (SEM)

In scanning electron microscope (SEM) a beam of electrons moves back and forth across the surface of a cell or tissue, creating detailed image of the 3D surface.

Q. What do you know about the discovery of cell?

Ans. **DISCOVERY OF CELL:**

A cell was discovered through observations. Discovery of cell is associated with the following facts.

1. Discovery of magnifying glass:

Observations were started with the discovery of magnifying glass.

2. Development of microscope:

Micro = Small; skopion = to see. In 1610 Galileo an Italian astronomer developed microscope to observe small organisms.

3. Invention of compound microscope and discovery of cell:

In 1665, Robert Hook made an improved microscope by combining lens, called compound microscope. He examined a slice of cork. He noticed honey comb like chambers which he named as cell (small room). He studied the cell wall only.

4. Discovery of nucleus:

In 1831 Robert Brown discovered a spherical body, the nucleus in the cells of orchids.

5. Discovery of plant cells:

In 1838 Schleiden a German botanist, proposed that all plants are made up of cells.

6. Discovery of animal cells:

In 1939 Theodor Schwann stated that all animals are made up of cells. He observed nuclei in all types of animal cells, but failed to observe cell wall in them.

7. Formation of new cells:

In 1858, Rudolf Virchow stated that new cells come only from other cells i.e. animal cells come from animal cell and plant cells from plant cell.

Q. What are the salient features of cell theory?

Ans. **SALIENT FEATURES OF CELL THEORY:**

The salient features of the cell theory are as under:

1. Composition of living organisms:

All living organisms including plants and animals are composed of cells. Among these some organisms are unicellular (one celled) and some are multicellular (more than one celled.)

2. Basic structural and functional unit:

Cell is the smallest, basic structural and functional unit of all organisms

3. Formation of new cells:

New cells are formed by the division of pre-existing cells.

Q. What is the structure of cell?

Ans. **STRUCTURE OF CELL**

A typical plant and animal cell consist of following parts.

1. CELL WALL:

Cell wall is the outermost boundary of plant cells, bacterial cells and fungal cells. It is not found in animal cell.

CHEMICAL COMPOSITION:

- It is secreted by the protoplasm of the plant cell.
- In plant cells it is mainly composed of cellulose and pectin.
- Bacterial cell wall is made up of protein and carbohydrate.
- Fungal cell wall is made up of fungal cellulose and chitin.

STRUCTURE:

- Ultra-microscope structure of cell wall shows that cellulose make the fibers which are arranged in crises cross manner/ These fibers are kept in their position by a cementing material called calcium pectate (Pectin)
- Thickness of cell wall is different in different plant cells.
- Cell wall is composed of three main layers,
 - **Middle lamella**
 - **Primary wall**
 - **Secondary wall**

MIDDLE LAMELLA:

Middle lamella is formed between the primary walls of two neighboring cells, it is cementing layer between two cells.

PRIMARY CELL WALL:

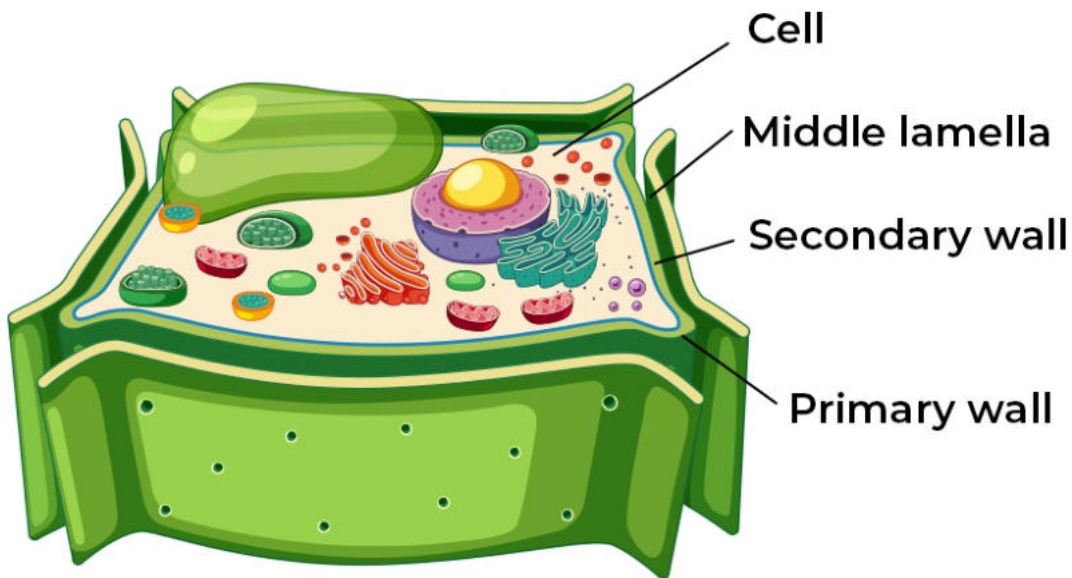
Primary wall is the first wall of plant cell. Chemically it is composed of cellulose and pectin sometimes, lignin.

SECONDARY WALL:

Secondary wall is formed by deposition of cellulose at the inside of primary wall.

FUNCTIONS:

- It provides protection and mechanical support to the cell.
- It acts like a skeletal framework of plants.
- It performs the function of transport of material (water and solute) from outside to inside, therefore it is permeable in nature.

CELL WALL STRUCTURE**2. CELL MEMBRANE:****DEFINITION:**

The cell membrane is the outer most living boundary of all cells.

BRIEF DESCRIPTION:

The cell membrane, also called the plasma membrane, physically separates the intracellular space (inside the cell) from the extracellular environment (outside the cell).

The cell membrane surrounds and protects the cytoplasm.

FUNCTION OF CELL MEMBRANE:

- It is semi-permeable membrane and regulates the flow of solution and materials in and out of the cell.
- It maintains the fluid concentration within the cell.
- Cell membrane takes in particular materials by the process of Endocytosis. It also intakes fluid material through the process of pinocytosis.

FLUID-MOSAIC MODEL OF STRUCTURE OF CELL MEMBRANE:

- According to fluid-mosaic model, by **S.J. Singer** and **G.I. Nicholson** (1972) plasma membrane composed of a phospholipids bi-layer as the core layer.

- The phospholipids layers lie parallel to each other.

- The lipid molecules are capable of rotating and moving within the membrane.

Protein molecules within the bi-layer float like ice bergs.

- Cell-membrane which is selectively permeable to ion, small molecules and even larger molecules allows the movement of substances in and out of the cells. · Cell-membrane performs many important functions for cell such as osmosis, diffusion, transport of nutrients and processes of ingestion and secretion.

1. DIFFUSION:

Diffusion is the movement of molecules from a region of high concentration to a region of low concentration. It is therefore said to occur down a concentration gradient.

2. OSMOSIS:

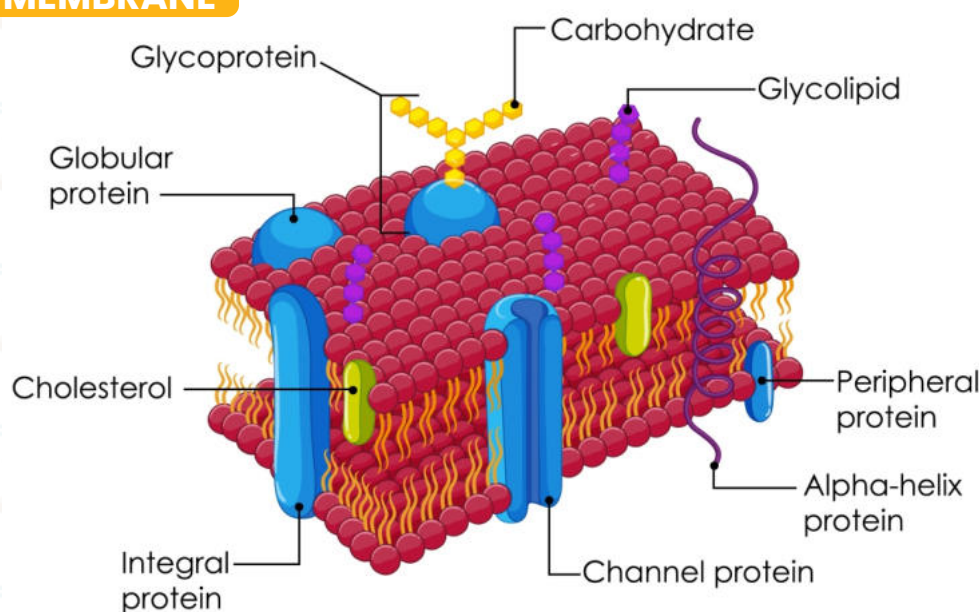
Osmosis is the movement of water molecules down a concentration gradient i.e. from dilute solution to concentrated solution.

IMPORTANCE OF OSMOSIS:

- Osmosis plays an important role in the survival of animal and plant cells as it is responsible to keep water and salt level balanced in blood and body,
- In plant cells, the absorption of water and its transportation occur due to osmosis.

In hypertonic conditions plant cell loses water and its cytoplasm shrinks resulting plasmolysis.

CELL MEMBRANE



TONICITY:

The term tonicity is defined as the relative concentration of solutes in the solutions being compared.

TYPES OF SOLUTION ON THE BASIS OF TONICITY:

On the basis of tonicity, there are three types of solutions.

- Hypertonic solution.
- Hypotonic solution.
- Isotonic solution.

HYPERTONIC SOLUTION:

A hypertonic solution is defined as a solution that has relatively more solute than its surroundings or compared solution.

HYPOTONIC SOLUTION:

A hypotonic solution has relatively less solute than its surroundings or compared solution.

ISOTONIC SOLUTION:

Isotonic solution is defined as a solution which has equal amount of solute to its surroundings or compared solution.

3. FACILITATED DIFFUSION:

Facilitated diffusion is the rapid exchange of specific molecules by the mean of special carrier protein.

EXPLANATION:

- Many molecules are transported across a membrane in the direction of low concentration by a carrier protein.
- Particles are taken up by the carrier proteins which change their shape. The change in shape causes the particles to be released on the other side of the membrane.

ACTIVE TRANSPORT:

It is the movement of substances against concentration gradient, from lower to higher concentration using input of energy.

- Active transport occurs when special carrier portion on the surface of a cell pick up particles from one side of the membrane and transport them to the other side.
- In biological systems the form in which this energy occurs is adenosine triphosphate (ATP).
- Movement of sodium and potassium ions across the cell membrane is an example of active transport.

Q. Write a detailed note on cell organelles. OR Describe a detailed note on cell organelles.

Ans. CELL ORGANELLES:

A cell is made up of multiple cell organelles, each one of them performs its specific function.

CYTOPLASM:

The protoplasm outside nucleus is called cytoplasm. It is the translucent fluid portion of the cell lying in between plasma membrane and nucleus.

CHEMICAL COMPOSITION:

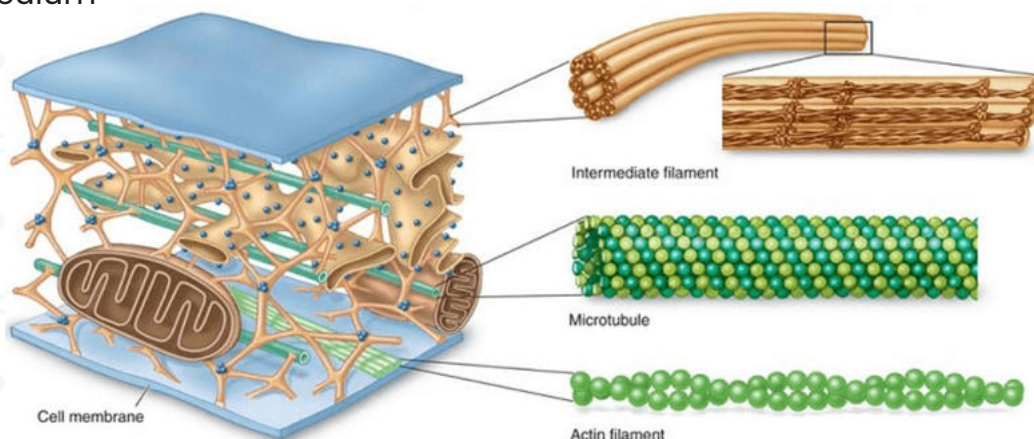
- Chemically cytoplasm is about 99% water.
- It consists of an aqueous ground substance called cytosol and granular portion called cytoplasmic organelles.
- It serves as store house of vital chemicals.

FUNCTIONS:

- Cytoplasm is a site of metabolic reactions like protein synthesis, glycolysis.
- Many reactions can occur at the same time in different regions of the cytoplasm.

SOME IMPORTANT CYTOPLASMIC ORGANELLES FOUND IN EUKARYOTIC CELLS.

1. Endoplasmic reticulum
2. Golgi complex
3. Mitochondria
4. Plastids
5. Centrioles
6. Ribosomes
7. Vacuole



CYTOSKELETON:

Cytoskeleton is a microscopic network of protein consists of microtubules and various filaments that are spread throughout the cytoplasm.

STRUCTURE AND COMPOSITION:

Microtubules of cytoskeleton are made of tubulin while filaments made up of active protein.

FUNCTIONS OF CYTOSKELETON:

- It provides structural support within cell.
- It provides mean of transport within cell.

NUCLEUS:

Nucleus was discovered by Robert Brown in 1831. It is an important and prominent structure lying inside the cell. In plant cell it is pushed towards cell membrane due to presence of large vacuole.

STRUCTURE:

Nucleus may be spherical or irregular in shape and it consists of following parts.

- Nuclear membrane
- Nucleoplasm
- Chromosomes

i. NUCLEAR MEMBRANE:

Nucleus is enveloped by a double membrane called nuclear membrane.

ii. NUCLEOPLASM:

Nucleus is filled with a gel like substance called nucleoplasm.

iii. CHROMOSOMES:

Chromatin network consists of thread like structures called chromosomes.

FUNCTIONS

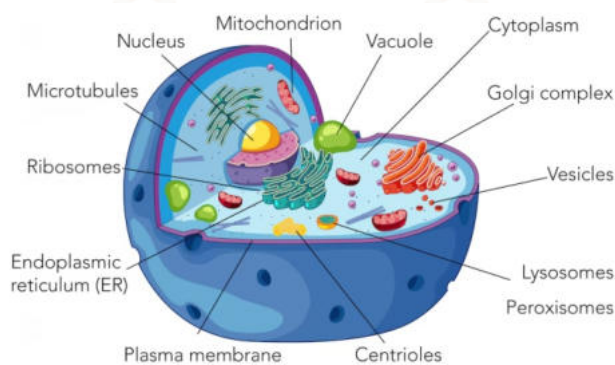
- Nucleus controls all the activities of cell.

DNA plays significant role in the inheritance of characters from parents to off springs.

MITOCHONDRIA:

DEFINITION:

Mitochondria are rod like or bean shaped cell organelles which consists of double membranes. They are energy producer.

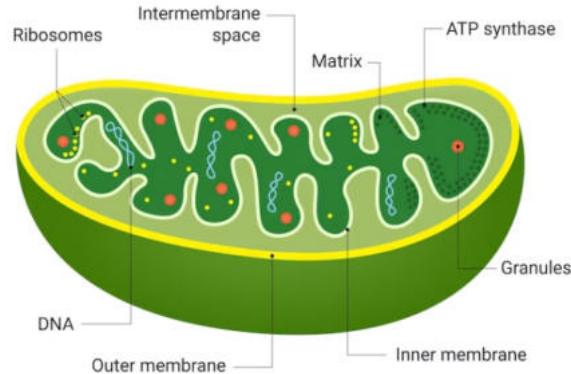


STRUCTURE:

Each mitochondrion is approximately about 0.2 to 1.0 μm in diameter and about 10 μm long. It consists of two thin membranes. The inner membrane is folded. These folding are called cristae while the fluid present inside is called **matrix**.

FUNCTION:

On the cristae (inner membrane) enzymes and co-enzymes are located which oxidize the food for the production of energy (respiration), therefore, they are called power house of the cell.

**ENDOPLASMIC RETICULUM:**

Endo=inside, plasma = protoplasm, reticulum = network

STRUCTURE:

It is a network of membrane channels or tubules extending throughout the cytoplasm. The channels seem to be in contact with plasma membrane as well as nuclear membrane.

COMPOSITION:

It is an elaborate tube-like system of lipoprotein.

TYPES OF ENDOPLASMIC RETICULUM:

There are two types of endoplasmic reticulum.

1. Rough endoplasmic reticulum (RER)

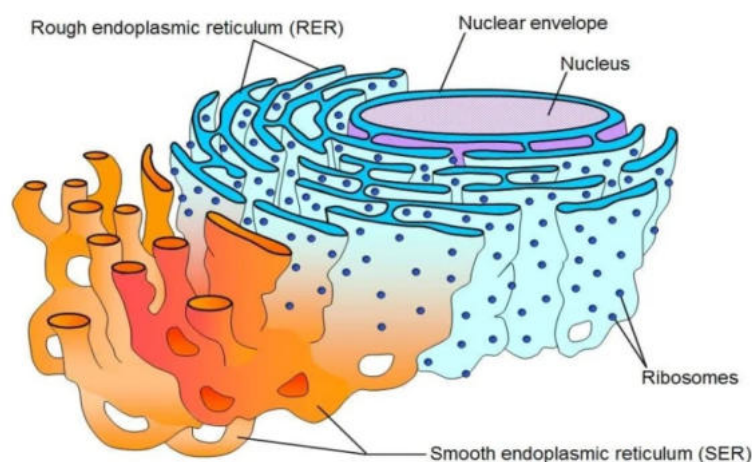
Rough endoplasmic reticulum is heavily coated with ribosomes on its outer surface which is involved in protein synthesis.

2. Smooth endoplasmic reticulum (SER)

Smooth endoplasmic reticulum is without ribosomes. It is found in steroid production cells.

FUNCTIONS:

- Endoplasmic reticulum controls the synthesis of protein and lipids.
- It transports RNA from nucleus to various organelles within the cell.
- It also provides mechanical support to the cell to maintain its shape.
- It detoxifies the harmful effects of drugs and chemicals



RIBOSOMES:**STRUCTURE:**

They are called ribosomes, because they are rich in Ribose Nucleic Acid (RNA). Each ribosome consists of two unequal subunits. They are found freely in cytoplasm as well as attached on the surface of rough endoplasmic reticulum.

FUNCTION:

They serve as sites where proteins are synthesized hence they are called protein factories of cell.

GOLGI COMPLEX:

They are discovered by Camilla Golgi and thus called Golgi complex or bodies.

STRUCTURE:

They are present inside cell in the form of small granules, fibers or flattened discs stacked together into flattened fluid filled sacs or vesicles containing carbohydrate, glycoproteins and enzymes.

FUNCTION:

- They are mainly concerned with the secretions e.g., polysaccharides.
- They manufacture certain macromolecules by itself.

VESICLES:

These are small, membrane-bound spherical sacs. Many vesicles are made in the Golgi body and the endoplasmic reticulum, or are made from parts of the cell membrane.

FUNCTIONS:

Vesicles facilitates the metabolism, transport and storage of molecules.

LYSOSOMES:

Lysosomes are small membrane-bordered structure that contains chemicals and enzymes necessary for digesting certain materials in the cell. Lysosomes are formed by the Golgi apparatus.

FUNCTIONS:

- Lysosomes are involved in digestion of certain cell material.
- Lysosomes are involved in breaking down organelles that have outlived their usefulness.

VACUOLES:

These are the fluid filled non-protoplasmic sacs (cavities) found in the cytoplasm of cells. Vacuole is surrounded by a selectively permeable boundary called tonoplast. The vacuole contains cell sap, in animal cells vacuoles are numerous and formed temporarily while in plant cells, a single, large and permanent vacuole is found.

FUNCTIONS:

- They store food material in the form of cell sap.
- They also sometimes play their role in plant defense.
- In animals' cells, they contain enzymes which help in digestion.
- They also play important role in process of hydrolysis, excretion of cellular waste.
- It also plays an important role in storage of water, organic and inorganic substances.

PLASTIDS:

Plastids are large cytoplasmic and major organelles found in cells of plants and algae.

STRUCTURE:

Each plastid is generally a spherical or ovoid body, bounded by membrane from outside.

TYPES OF PLASTIDS:

The contain different type of pigments on the basis of which they are classified into following types.

- Chloroplast (Green-colored plastids found in plants and algae)
- Chromoplast (Contain-red, orange or yellow pigments and are common in ripening fruit, flowers or autumn leaves.)
- Leucoplast (Color less plastids)

CHLOROPLASTS (GREEN PLASTIDS)

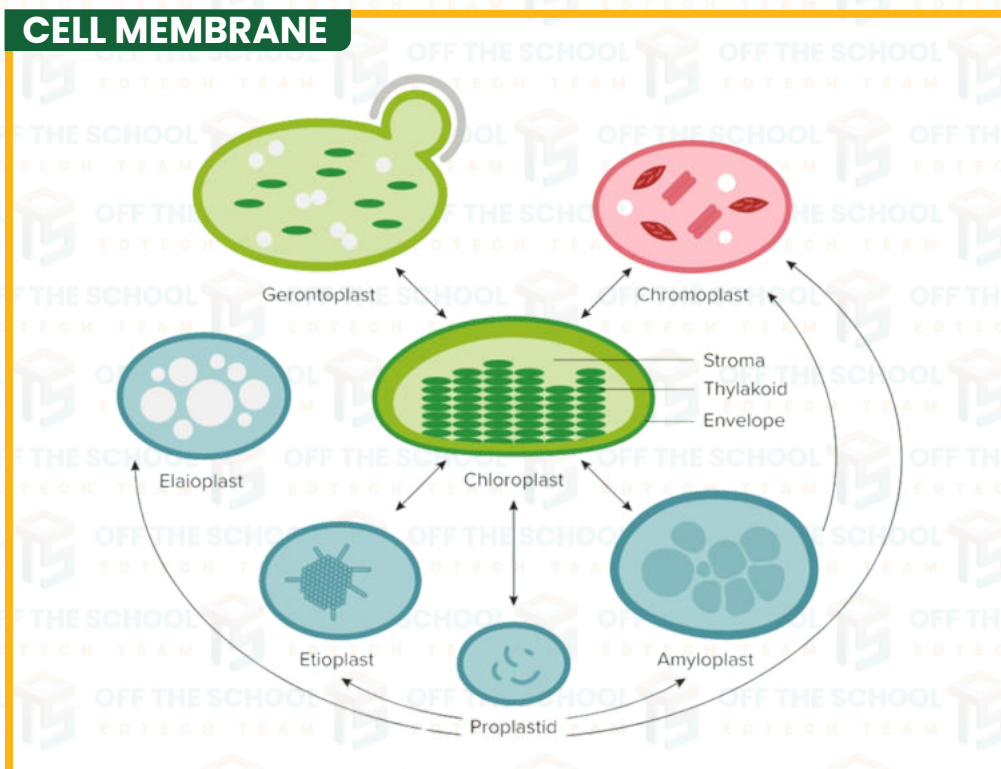
Chloroplasts are double membrane, bean shaped cytoplasmic organelle which contain chlorophyll and found in all green parts of plants.

STRUCTURE:

- The chloroplast is double-membrane organelle.
- Within the membrane is a gel like substance called stroma.
- Stroma contains enzymes for photosynthesis and suspended grana (singular=granum).
- Each granum is a stack of thylakoid discs.
- The chlorophyll molecules (green pigments) are found on the surface of the thylakoid discs.

FUNCTIONS:

- They absorb light energy with the help of chlorophyll for photosynthesis.
- They manufacture carbohydrates by the process of photosynthesis



Q. Briefly describe variations in cell size?

Ans. **VARIATIONS IN CELL SIZE:**

Cells vary greatly in size. The smallest cells are bacteria called Mycoplasma, with diameter between $0.1\text{ }\mu\text{m}$ to $1.0\text{ }\mu\text{m}$. The bulkiest cells are bird eggs, and the longest cells are some muscle cells and nerve cells. Most cells lie between these extremes.

IMPORTANCE OF CELL SIZES OR RELATION OF CELL SIZE WITH FUNCTION:

- Cell size and shape are related to cell function.
- Bird eggs are bulky because they contain a large amount of nutrient for the developing young.

- Long muscle cells are efficient in pulling different body parts together.
 - Lengthy nerve cells can transmit messages between different parts of body.
- Small cell size also has many benefits, for example human red blood cells are only 8 μm in diameter and therefore can move through our tiniest blood vessels i.e., capillaries.

VARIATION IN SURFACE TO VOLUME RATIO:

Most cells are small in size. In relation of their volumes, large cells have less surface area as compared to small cells.

IMPORTANCE OF SURFACE TO VOLUME RATIO:

Need of nutrients and rate of waste production are directly proportional to cell volume. Cell takes up nutrients and excretes wastes through its surface cell membrane. So, a large volume cell demands large surface area.

Q. Define tissue. Explain in detail animal tissues.

Ans **TISSUE:**

DEFINITION:

A group of similar type of cells organize together in a group called tissue.

ANIMAL TISSUES:

Animals are made up of four basic types of tissues

- Epithelial tissue
- Connective tissue
- Muscular tissue
- Nervous tissue

1. EPITHELIAL TISSUE:

Epithelial tissue covers the surface of the body, lines the space inside the body and forms glands.

LOCATION:

TYPES OF EPITHELIAL TISSUE:

The simplest classification of these tissues is based on the number of cell layers.

- Simple epithelial tissue
- Striated epithelial tissue

1. SIMPLE EPITHELIAL TISSUE:

When the epithelium is composed of single layer of cells, it is called simple epithelial tissue.

TYPES OF SIMPLE EPITHELIAL TISSUE:

Simple epithelial tissue can be classified into following three types.

a) Simple squamous epithelium:

- A simple squamous epithelium is a single layer of flat cells.
- Simple squamous epithelium is found in the alveoli of lungs.
- It is important for exchange of gases between blood and lungs.

b) Simple cuboidal epithelium:

- Simple cuboidal epithelium consists of single layer cube shaped cells that are tall as they are wide.
- Simple cuboidal epithelia line the lumen of collecting duct in the kidney, pancreas and salivary glands and present in thyroid gland around the follicle.
- Important functions of the simple cuboidal epithelium are secretion and absorption

c) Simple columnar epithelium:

Simple columnar epithelia are tissues made of a single layer of long column shaped epithelial cells. It is found in the female reproductive system and in the digestive tract. This tissue is associated with absorption and secretion

2 STRATIFIED EPITHELIAL TISSUES:

Stratified epithelial consists of more than one layer of cells and only one layer is in direct contact with the basement membrane.

TYPES OF STRATIFIED EPITHELIAL TISSUE:

Stratified epithelial tissue can be classified into following types

a) Stratified cuboidal epithelia:

Stratified cuboidal epithelia are found surrounding the ducts of many glands, including mammary glands in the breast and salivary glands in the mouth.

b) Stratified columnar epithelia:

Stratified columnar epithelia are rare, found predominantly in some organs of the reproductive system.

c) Transitional epithelia:

Transitional epithelia are a special subset of stratified epithelia. They are exclusively found in the excretory system.

2 CONNECTIVE TISSUE

The tissue which connects or bind the different types of cells called connective tissue. These are made up cells, fibers and a gel-like substance. These tissues are widely found through the body between tissue and organs and also form sheaths around the organs.

FUNCTIONS:

Connective tissues serve many purposes in the body

- Connective tissues bind structure together.
- It provides support and protection. Fills space, stores fat, and form blood cells.
- It also provides the source cells for muscular and skeletal cells.
- It can repair the damage and injured tissues.

3 MUSCULAR TISSUE:

This tissue consists of muscles fibers. Each fiber consists of elongated cell containing nucleus.

Functions

- It has ability to contract and relax.
- It causes movement of the body, body parts and material through internal organs

4 NERVOUS TISSUES

Nervous tissues consist of nerve cells called neurons or nerve cell. Each nerve cell is specialized cell consisting of three parts. Nervous system consisting of nervous tissues.

Function

- It is responsible for coordinating and controlling many body activities.
- It transmits information to other cells in the form of electrical nerve impulses.
- It stimulates muscle contraction, creates an awareness of the environment, and plays a major role in emotions, memory and reasoning

TYPES OF CONNECTIVE TISSUE:

1 CARTILAGE

- It is a type of supporting connective tissue,
- It is a dense connective tissue,
- Cartilage protects and covers the ends of long bones at the joint and nerves, and is a structural component of the rib cage, the ear, the nose, the bronchial tubes, the intervertebral discs, and many other body components.

BONE

Bone is a type of connective tissue.

- It is a honey comb-like matrix internally, which helps to give the bone rigidity.
- Bone tissue is made up of different types of bone cells mainly osteoblast and osteocytes.
- Bones protect vital organs, enable movement, and provide attachment sites for muscles and tendons

2 ADIPOSE:

Adipose tissue is a specialized connective tissue consisting of lipid-rich cell. The main function of adipose tissue to provide cushions and store energy in the form of lipids (fat)

3 BLOOD:

Blood is a specialized fluid connective tissue. Blood circulates within the body and serves as a medium of transport and exchange of material.

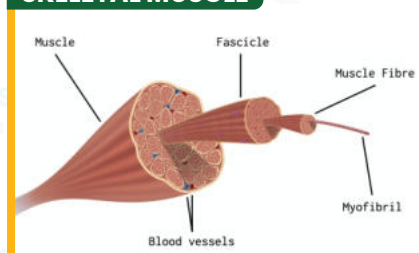
TYPES OF MUSCLE TISSUES: They are following types of muscle tissues.

- Skeletal muscle
- Cardiac muscle
- Smooth muscle

i Skeletal Muscle

- These types of muscle are attached with bones.
- The cells are striated (striped) contain many nuclei.
- They are voluntary in nature
- They are responsible for the movement of bones.

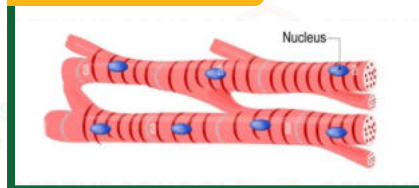
SKELETAL MUSCLE



ii Cardiac Muscle

- These muscles are present in the wall of heart.
- Their cells are striated but with a single nucleus cell.
- They are involuntary in nature.
- They produce heart beat

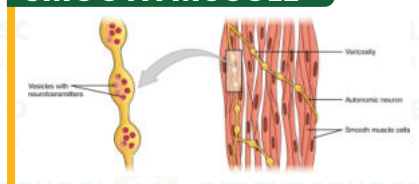
CARDIAC MUSCLE



iii Smooth Muscle

- Smooth muscles are found in the wall of alimentary canal, urinary bladder, blood vessels etc.
- They contain smooth (non-striated) cells, each with a single nucleus. They are involuntary in nature.
- They are responsible for the movement of substances.

SMOOTH MUSCLE



Q. Briefly describe plant tissues?

Ans. **PLANT TISSUES**

A plant tissue is defined as a group of similar cells to perform similar function and exhibit similar properties for example photosynthesis, transport etc.

CLASSIFICATION OF PLANT TISSUES:

There are two major categories of tissues in plants i.e. Meristematic tissues and Permanent tissues.

1. MERISTEMATIC TISSUES:

DEFINITION:

A meristematic tissue is the plant tissue that has the ability to divide actively throughout its life.

LOCATION:

They are found at the tips (apex) and sides of roots, shoots and stem.

SALIENT FEATURES:

- These tissues are connected of cells, which have the ability to divide.

- The cells are thin walled, have large nucleus and number of small vacuoles.
- Usually they do not have inter-cellular spaces, so the cells are arranged compactly.

TYPES:

These are two main types of meristematic tissues in plants.

- Apical meristematic.
- Lateral meristematic.

i. APICAL MERISTEMATIC:

- Apical meristematic tissues are present at the apex of roots and stems.
- According to their position they are apical meristematic.
- Stem and root increase in length by the division of cells of these tissues, this type of growth is called primary growth.

ii. LATERAL MERISTEMATIC:

- Lateral meristematic are located on the lateral sides of roots and shoot.
- Lateral meristematic are responsible for increase in girth (breadth or thickness) of plant parts. This growth is called **secondary growth**.

2. PERMANENT TISSUES:

DEFINITION:

Plant tissue that has completed its growth and differentiation and is usually incapable of meristematic activity.

ORIGIN:

They originate from primary meristem.

SALIENT FEATURES:

- The cells of these tissues do not have the ability to divide.
- They may have intercellular spaces in between cells.
- They have differentiated cells to carry out specific functions.

TYPES OF PERMANENT TISSUES:

There are two types of permanent tissues.

- Simple permanent tissue.
- Complex or compound tissue.

1. SIMPLE PERMANENT TISSUE:

The simple tissue is made up of one type of cells forming a uniform or homogeneous system of cell.

TYPES OF SIMPLE PERMANENT TISSUE:

Simple tissues may further divide into following types.

- Epidermal tissue.
- Ground tissue.
- Supporting tissue.

1. EPIDERMAL TISSUE:

The tissue which forms the outer protective covering of the plant and is represented, in the primary plant body, by epidermis is called dermal (epidermal) tissue.

FEATURES:

- Epidermal tissues are composed of a single layer of cells.
- They cover plant body.

FUNCTIONS:

- They act as a barrier between environment and internal plant tissues.
- In roots, they are also responsible for the absorption of water and minerals.

- On stem and leaves they secrete cutin (the coating of cutin is called cuticle) which prevent evaporation.
- Epidermal tissues also have some specialized structure that perform specific functions; **for example**, root hairs and stomata.

2. GROUND TISSUE:

DEFINITION:

A ground tissue arises from the ground meristem, it fills in the soft parts of the plants, such as cortex, pith, pericycle, etc.

SALIENT FEATURES:

- These tissues are made up of parenchyma cells.
- They are spherical but flat at point of contact.
- They have thin primary cell wall.
- They have large vacuoles for storage of food.
- In leaves, they are known as mesophyll.

FUNCTIONS:

- Their primary function is the storage of food.
- They are also responsible for photosynthesis.
- These tissues also provide structural support to plant.

3. SUPPORTING TISSUES:

DEFINITION:

The tissues which provide strength, flexibility and structural support to plant are called as supporting tissues.

FUNCTIONS:

They provide structural support and rigidity to plant.

TYPES:

They are further classified into two types.

- Collenchyma.
- Sclerenchyma.

1. COLLENCHYMA:

Collenchyma is a tissue composed of living elongated cells having unevenly thickened primary cell walls.

LOCATION:

They are mostly found in cortex of young stems, midribs of leaves and in petals of flowers.

FUNCTION:

These tissues provide structural support to plants without rejecting growth due to their relatively soft cell wall.

2. SCLERENCHYMA:

The sclerenchyma consists of thick and rigid secondary cell walls, often unified.

LOCATION:

They are often found in xylem and hard fruit coats etc.

SALIENT FEATURES:

- It consists of dead cells.
- The cell walls are lignified.

2. COMPOUND OR COMPLEX TISSUE:

DEFINITION:

A plant tissue composed of more than one type of cell is called a compound or complex tissue.

EXAMPLES OR TYPES OF COMPOUND TISSUES:

Xylem and phloem tissues, found only in vascular plants, are example of composed tissues.

- Xylem.
- Phloem.

1. XYLEM:**DEFINITION:**

The tissue that functions to transport water and minerals from roots to aerial parts is called xylem.

LOCATION:

In dicot they are located inside the phloem.

SALIENT FEATURE:

- Due to the presence of lignin, the secondary walls of its cells are thick and rigid with overlapping ends.

STRUCTURE:

It is composed of following elements.

- **Tracheid's** are non-living, long, narrow and thin cells with tapered ends.
- **Vessels** are non-living, short, wide and stacked end forming a tube-like structure with perforated walls.

FUNCTIONS:

- They perform the function of transport of water and minerals from roots to aerial parts of plant. (Ascent of sap)
- They also provide structural support to the plant.

2. PHLOEM:**DEFINITION**

The tissue that transports nutrients from leaves to various parts of plant is called phloem. Phloem is a complex, food conducting tissue in vascular plants.

LOCATION:

In dicot they are located outside the xylem.

STRUCTURE:

Phloem is composed of four elements:

- **Sieve tubes** are elongated living cells with perforated end walls, which do not possess nucleus and are main conducting cells.
- **Companion cells** which are closely associated with sieve tubes and control the sieve tubes.
- Phloem parenchyma which is mainly concerned with storage of organic matter, tannins etc.
- Phloem fibers are elongated cells with lignified walls.

FUNCTIONS:

They transport food and other metabolic products from leaves to various parts of plant.

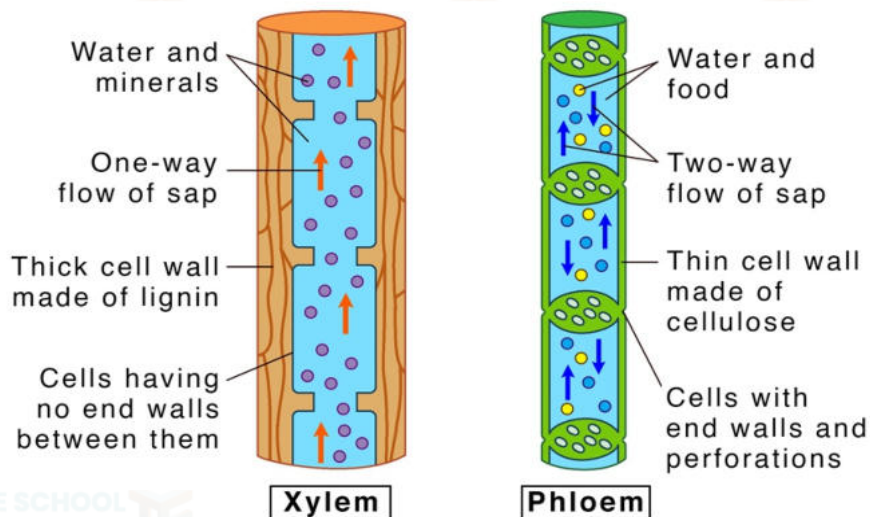
**Fun Fact**

The Average Person's Body Contains Enough DNA To Stretch From The Sun To Pluto And Back



17 Time!

XYLEM & PHLOEM



SCIENTIFIC REASONS

Q.1. Why iodine used to stain the onion peel?

Ans. Iodine is used to stain the onion peel because it reacts with starch and turns blue-black, causing the cell membrane and nucleus to be more visible.

Q.2. Why cell membrane is semi-permeable in nature?

Ans. The cell membrane exists in nature because it is responsible for performing many important functions within the cell, such as osmosis, diffusion, transport of nutrients into the cells, and the process of ingestion and secretion.

Q.3. Why cell is considered as the structural and functional unit of living things?

Ans. Cells are considered to be the structural and functional unit because the body of every living organism is made up of cells or cell groups. It is said to be the functional unit of life because the cell is responsible for performing

Fun Fact

More Than **100000**
Chemical Reactions
Occur Every Second In Our
Brain.

